

Dr. Louise Fouqueau

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Summary Statement

I am an eco-evolutionary biologist bringing together empirical (fieldwork, data, experiments) and theoretical approaches (simulation and mathematical models) to study general questions concerning the effect of spatial structure (e.g., habitat fragmentation) on the adaptive potential of populations and their capacity to persist in the face of the accumulation of deleterious mutations. During my PhD, I was interested in the unique characteristics of populations located at species' range margins and/or expansion fronts, focusing specifically on the evolution of clonality and its consequences. During my postdoctoral research, I have delved further into studying the effects population structure and the genetic architecture of traits under selection, as both of these features have a significant impact on eco-evolutionary feedback.

Education

- 2017-2021 PhD at the Biological Station of Roscoff (France).
- 2016-2017 Dual degree programme combining a Master's in Biology, Ecology and Evolution from the University of Montpellier and a Master's from AgroParisTech.
- 2015-2016 Full year internship.
- 2013-2015 1st and 2nd year at AgroParisTech. Major in "Management and Engineering of the Environment".
- 2012-2013 Diderot University (Paris, France), major in Life Sciences.
- 2010-2011 Preparatory classes (BCPST) in Lycée Jean Baptiste Say (Paris, France).

Research and Teaching experience

Postdoctoral Research, ISTA (Austria)

September 2022 –

1) Exploration of the effect of population structure on species' ability to persist in regard to the accumulation of deleterious mutations and environmental change, effect of reproductive systems and genetic architecture on these outcomes; 2) Field work on *Antirrhinum majus* in the Spanish Pyrenees; 3) Teaching.

Fieldwork (Survey, Sampling, Organization), Statistical & Data analysis (R), Mathematical models (Mathematica), Simulations (C++)

Postdoctoral Research, Biological Station of Roscoff (France).

January – May 2022

Progress on PhD projects. Grant application.

PhD, Biological Station of Roscoff (France).

November 2017 – December 2021

Supervised by Myriam Valero and Denis Roze.

1) Exploration of the influence of historical and contemporary factors on the geographic distribution of genetic diversity in *Laminaria digitata*; 2) Exploration of the evolution of clonality in species' range limits and its consequences on range expansion; 3) Teaching.

Fieldwork (Survey, Sampling, Determination, Organization), Experimental Work (Common Garden Experiment, Cultivation), Laboratory work (DNA extraction), Statistical & Data analysis, Mathematical models, Simulations

Research internship, CEFÉ, Montpellier (France).

February – July 2017

Supervised by Cyrille Violle and François Vasseur.

Examination of the physiological dominance model on the emergence of heterosis in two fitness-related traits (growth rate and fruit number) measured on 450 hybrids of in *Arabidopsis thaliana*.

Statistical Analyses, Bioinformatics.

Research internship, Kyushu University, Fukuoka (Japan).

April – July 2016

Supervised by Yahara Tetsukazu.

Examination of the variation in plant diversity under deer (*Cervus nippon yakushimae*) grazing pressure on the island of Yakushima (Japan). Illumina sequencing of a hundred plant species in order to identify the eaten species from deer feces.

Fieldwork (Survey, Determination, Quadrats), Laboratory work (Sample preparation, DNA extraction)

Research internship, CRIOBE, Moorea (French Polynesia).

September 2015 – March 2016

Supervised by Laetitia Hédouin and Antoine Puisay.

1) Exploration of the resistance of young coral stages in the face of increasing seawater temperature (*Acropora* and *Pocillopora* species); 2) Study on the correlation between the fluorescence of *Pocillopora damicornis* larvae and their survival and recruitment abilities.

Fieldwork (Survey of transplants, Sampling), Experimental work, Data analysis.

Research internship, CEFÉ, Montpellier (France).

June – August 2015

Supervised by Luis-Miguel Chevin.

Demographical dynamics of populations submitted to various environmental changes. Quantitative genetic approach.

Simulations.

Conference organisation

Member of the STN “Dispersal & Mating Systems”, and organizer of the Seminar Series (2024 - today)

Main organizer of Nick Barton’s 70th anniversary conference (2024-2025)

Main organizer of the symposium “eco-evolutionary dynamics in variable environments” (ESEB, 2022)

Main organizer of the 26th edition of EMPSEB (2021)

Grants & Awards

Three years postdoctoral grant, financed by the NOMIS foundation (2022)

Best presentation award at the Young Researchers Day (2020)

Best presentation award at the conference of the French Phycological Society (2020)

Mobility grant TRANSHUMANCE from the Doctoral School (ED227 MNHN-SU) (2019)

Supervision

Main Supervision

Martin Gbúr (Rotation) 14/04/2025 – 30/06/2025

Infinitesimal model on the emergence of range limits. Co-supervised by Nick Barton (ISTA).

Simon Fourmann (Master) 01/03/2024 – 30/06/2024

Probability and fixation time of a deleterious mutations in a structured populations with self-fertilization. Co-supervised by Himani Sachdeva (Uni. Wien)

Léa Cardinal (Bachelor) 12/04/2021 – 18/06/2021

Reproductive systems in central and marginal populations of *L. digitata*. Co-supervised by Myriam Valero (CNRS).

Sébastien Guillermin (Master) 20/04/2020 – 30/06/2020

Temporal analysis of genetic diversity of *L. digitata*. Co-supervised by Myriam Valero (CNRS).

Magdalena Mendez (Technician) 01/10/2018 – 31/03/2019

Optimization of *L. digitata* cultivation. Co-supervised by Myriam Valero (CNRS).

Aurélien Baud (Technician) 01/07/2018 – 31/08/2019

Genotyping and species identification. Co-supervised by Myriam Valero (CNRS).

Co-Supervision

Danmen Wang & Runtian Wu (Master) 30/09/2025 – 31/01/2026

Distribution of deleterious mutations along the genome. Main supervisor Diala Abu Awad (Uni. Saclay)

Axel Fichou (Master) 20/04/2020 – 29/05/2020

Analyses of demographical survey in two populations of *L. digitata*. Main supervisor Myriam Valero (CNRS).

Kseniia Kudiakhova (Rotation) 15/09/2019 – 15/12/2019

Models on the evolution of sexual reproduction. Main supervisor Nick Barton (ISTA)

Paul Martel (Bachelor) 05/01/2019 – 31/03/2019

Optimization of *L. digitata* cultivation. Main supervisor Emeline Creis (Station Biologique de Roscoff)

Editorial work & Review

Reviews for *Evolution*, *Heredity*, *Journal of Evolutionary Biology*, *Molecular Ecology*, *Oikos*, *Proceedings of the National Academy of Sciences*, *Genetics*, *Integrative and Comparative Biology*...

Invited Editor in *Journal of Evolutionary Biology* (2022-2024).

Selected talks and conferences (only until 2023)

EvoVienna Seminar (September 2026). Invited Speaker

Internal Seminar. iEES Paris (June 2026). Invited Speaker.

ISTA Nomis fellows (March 2026). Invited Speaker.

Internal Seminar. ISEM (February 2026). Invited Speaker.

PopGroup Lille. (January 2026). Speaker, Chairman.

Internal Seminar. UNIL (October 2025). Invited Speaker.

BrainFoods. ISTA (October 2025). Invited Speaker.

Workshop “Probability Meets Biology – Berlin Edition” (November 2024). Invited Speaker.

Internal seminar ECOBIO, University of Rennes (September 2023). Invited Speaker.

University of Basel. (June 2023) Invited Speaker.

Internal Seminar. Mathematical Department of the University of Vienna (May 2023). Invited Speaker.

Internal Seminar. Botanical Department of the University of Vienna (April 2023). Invited Speaker.

Languages

French (native), Japanese (native), English (fluent), Spanish (good), German (A2, improving)

Informatics: R, SIG, Bash, C++, Mathematica, LaTeX

Other

Association: Klimacamp (2022-2024), founding member of “Gasièr.e.s de la Baie”, volunteer for “Moyens du Bord” (2017-2021) and “Bretagne vivante” (2017-2022), member of the administrative board and grant application group for “Reder Mor” (2020-2022), treasurer for “Association des Jeunes Chercheurs” (2018).

Non-Academic work: Wallon-Marcadeau Hut (2011-2014), market gardening.

Sports: Ashtanga yoga, hiking/trekking, bike-packing, trail running, mountaineering, climbing, badminton.